

CareerHub

Advanced Web Programming – Project Phase 2

Student	Yehya Naser Dine
Course	CSCI426 – Advanced Web Programming
Project Type	Full-Stack Web Application
Backend	Node.js and Express
Database	MySQL / XAMPP

GitHub: <https://github.com/82330148-byte/careerhub-react>

Frontend: <https://careerhub-react-o7sv-r21nhmt68-power-gym-react.vercel.app/>

1. Project Description

CareerHub is a job-search and recruitment platform that connects users with available jobs. Users can create accounts, log in, browse jobs, view job details, save jobs, apply by uploading a CV, update their profile, and send contact messages. An administrator can manage job records and application status through an admin dashboard.

2. Technologies Used

Layer	Technology
Frontend	React, Vite, React Router, Axios
Backend	Node.js, Express.js
Database	MySQL using XAMPP and phpMyAdmin
Authentication	bcryptjs password hashing
File Upload	Multer
Version Control	Git and GitHub
Deployment	Vercel for the React frontend

3. Main Functions

- User registration and login with password hashing.
- Job listing, job details, save/remove jobs, and job applications.
- CV file upload using Multer and FormData.
- Profile reading and updating from the MySQL database.
- Contact form messages stored in MySQL.
- Admin CRUD operations for jobs and application-status updates.
- Data validation, duplicate checks, and API error handling.

4. Database Design

The database is named **careerhub_db**. It stores related entities using foreign keys. The main tables are users, jobs, saved_jobs, applications, and contact_messages. The relationships allow a user to save multiple jobs and submit applications for different jobs.

Table	Purpose / Relationship
users	Stores user account, email, password hash, and role.
jobs	Stores available job records.
saved_jobs	Links users and jobs using user_id and job_id.
applications	Links a user to a job and stores CV path and status.

Table	Purpose / Relationship
contact_messages	Stores messages sent through the contact page.

5. Key Code Snippets

MySQL connection:

```
const db = mysql.createConnection({
  host: "localhost",
  user: "root",
  password: "",
  database: "careerhub_db"
});
```

Read jobs from the database:

```
app.get("/jobs", (req, res) => {
  db.query("SELECT * FROM jobs ORDER BY created_at DESC", (error, data) => {
    if (error) return res.status(500).json({ message: error.message });
    res.json(data);
  });
});
```

Create a new user with a hashed password:

```
const hashedPassword = await bcrypt.hash(password, 10);
db.query(
  "INSERT INTO users (full_name, email, password, role) VALUES (?, ?, ?, 'user')",
  [full_name, email, hashedPassword], callback
);
```

The application also implements POST, PUT, and DELETE routes for complete CRUD operations.

6. Application Screenshots

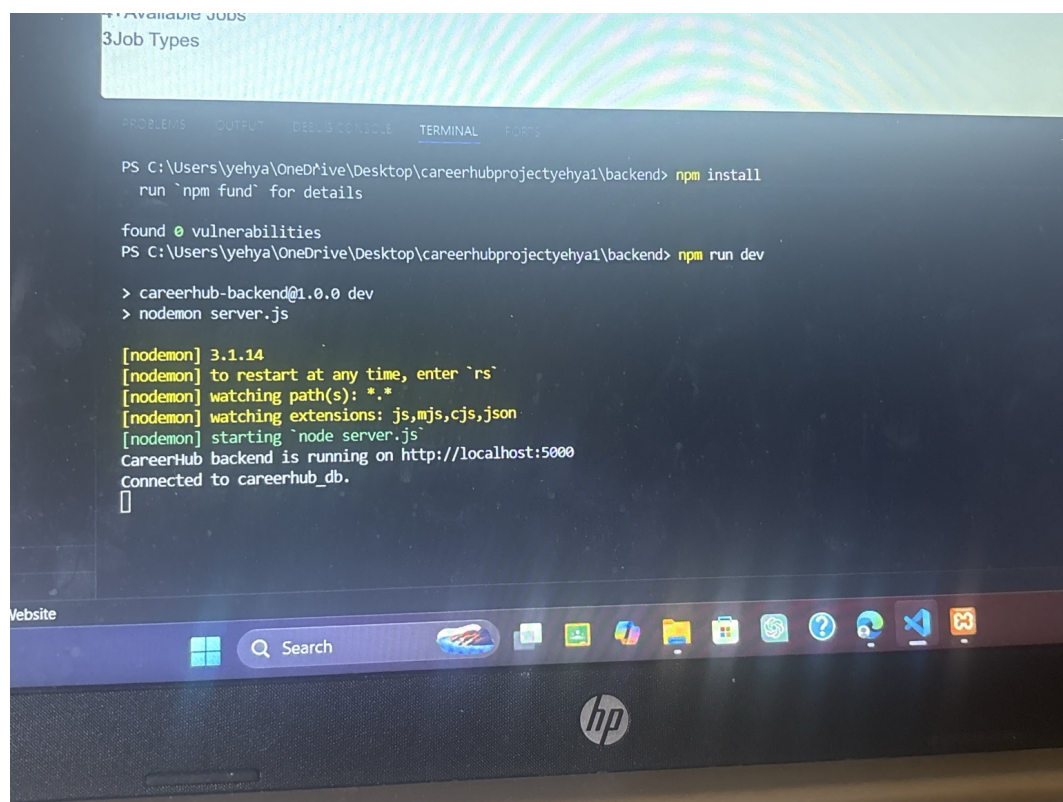


Figure 1. Node.js backend running and connected to careerhub_db.

7. Setup and Testing

The project is tested locally using XAMPP. Apache and MySQL are started, the SQL file is imported into phpMyAdmin, the backend is started on localhost:5000, and the frontend is started on localhost:5173. The jobs API and all main pages were tested, including registration, login, profile, saved jobs, applications, contact, and admin features.

Backend commands:

```
cd backend  
npm install  
npm run dev
```

Frontend commands:

```
cd frontend  
npm install  
npm run dev
```

8. Conclusion

CareerHub meets the main Phase 2 requirements by using Node.js as a backend, MySQL database integration, login and signup, related database entities, validation, error handling, CRUD operations, Git/GitHub version control, documentation, and an admin panel. The application demonstrates client-server communication between React, Express, and MySQL.

9. Future Scope

Future improvements can include online backend and database hosting, email notifications, job search filters, pagination, JWT-based authentication, cloud CV storage, and stronger role-based authorization.

10. Contribution Statement

This project was completed individually by **Yehya Naser Dine**. I was responsible for frontend development, backend development, database design and integration, testing, documentation, version control, and frontend deployment.